

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Distributed Systems Aug 2025 to May 2026

- Reduced rebuffering by **19%** versus a fixed-rate baseline by building a **Docker**, **WebTransport**, and **Media-over-QUIC** testbed and running controlled adaptation experiments across network and protocol configurations.
- Cut experiment setup effort by about **55%** by automating orchestration, configuration, **telemetry** collection, and analysis, making large experiment sweeps reproducible.
- Built and exercised **14 interoperability scenarios** covering networking, dependency, and **failure recovery** behavior in a distributed real-time system.
- Boosted malware classification to **80.2%** accuracy across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.

Turito | Software Engineering Intern, Systems Performance Jun 2022 to Oct 2022

- Slashed p95 **CPU** by **22%** by profiling high-frequency media paths and removing excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures by **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.
- Prototyped a **Rust** + **QUIC** media path that supported about **8,000** lightweight subscribers with about **24%** lower startup time.
- Lowered network egress by **14%** by removing redundant media-path processing and simplifying routing and payload handling.
- Reduced production diagnosis time by **35%** by adding structured **metrics**, runtime-state visibility, and targeted logs.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Surfaced backend fixes that raised equivalent-load signaling throughput by about **31%**.

TECHNICAL SKILLS

Languages: C++17, Rust, Python, C, SQL, TypeScript

Systems & Performance: AVX2/SIMD, multithreading, lock contention analysis, profiling, memory layout, Linux, CMake, pybind11

Distributed & Networking: Tokio, QUIC, WebTransport, WebRTC, RTP/RTCP, WebSockets, gRPC, Kafka, Docker

Data & Search: HNSW, vector search, PostgreSQL, SQLite, Redis, Hadoop, Hive

PROJECTS

Proxima, Vector Search Engine | C++17, AVX2/SIMD, HNSW, pybind11

- Built an HNSW vector search engine from scratch that reaches **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** faster than hnsplib and **2.5x** faster than FAISS on 100K vectors.
- Cut traversal memory by **75%** (51 MB to 12.8 MB) with **int8** traversal on AVX2 kernels plus exact float re-ranking, and cache-aware node reordering.

TCP Sliding Window Simulator | Python, raw sockets, Linux

- Implemented handshake, sliding window, cumulative ACKs, and retransmission over raw sockets with live packet-state visualization.

OPEN SOURCE

- fission/fission:** 3 merged PRs, Kubernetes event and KEDA object capture in support dumps, removed duplicate package enumeration
- google/osv-scalibr:** upstream change to Google's software composition analysis library (open PR, awaiting review)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Distributed Computing, Advanced Algorithms, Database Systems, Computer Networks, GPU Programming, Machine Learning

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)